Computer Networks - Homework 1

Task Description

Given a pcap file, analyze it and answer the following questions:

1. What is the IP Address Requested by the client via DHCP?
2. What is the IPv6 Address of the NTP server?
3. What is the name of the authoritative name server for the queried domain?
4. What is the CDP protocol port for host CCNP-LAB-S2?
5. What IOS version is running on CCNP-LAB-S2?
6. When was the NVRAM config last updated?

I included the packet numbers in the file between parentheses for easier reference.

Question 1

Answer: `192.168.30.11`

First, I filtered by **DHCP**.

dhcp

No.	Time	Source	Destination	Protocol	Length	Info
1254	121.772905	0.0.0.0	255.255.255.255	DHCP	346	DHCP Request - Transaction ID 0x5f511e61
1255	121.782156	192.168.30.1	255.255.255.255	DHCP	346	DHCP NAK - Transaction ID 0x5f511e61
1256	121.805157	0.0.0.0	255.255.255.255	DHCP	346	DHCP Discover - Transaction ID 0x96a1041e
1257	121.810908	192.168.30.1	192.168.30.11	DHCP	361	DHCP Offer - Transaction ID 0x96a1041e
1258	121.821659	0.0.0.0	255.255.255.255	DHCP	346	DHCP Request - Transaction ID 0x96a1041e
1259	121.827910	192.168.30.1	192.168.30.11	DHCP	361	DHCP ACK - Transaction ID 0x96a1041e

The client first requested the IP 192.168.20.11(1254), but was denied(Received a NAK response, packet 1255)

After sending a DHCP **Discover**(1256) and receiving an **Offer**(1257) for the IP 192.168.30.11, the client requested IP **192.168.30.11**(1258), which was acknowledged(1259)

Question 2

Answer: `2003:51:6012:110::dcf7:123`

First, I filtered by **NTP**.

ntp

No.	Time	Source	Destination	Protocol	Length	Info
157	13.806387	192.168.121.40	212.224.120.164	NTP	94	NTP Version 3, client
158	13.808394	212.224.120.164	192.168.121.40	NTP	94	NTP Version 3, server
187	15.802944	192.168.121.40	78.46.107.140	NTP	94	NTP Version 3, client
188	15.809945	78.46.107.140	192.168.121.40	NTP	94	NTP Version 3, server
459	36.808209	192.168.121.40	148.251.154.36	NTP	94	NTP Version 3, client
460	36.814964	148.251.154.36	192.168.121.40	NTP	94	NTP Version 3, server
2918	286.367467	2003:51:6012:121::10	2003:51:6012:110::dcf7:123	NTP	114	NTP Version 4, client
2919	286.368969	2003:51:6012:110::d…	2003:51:6012:121::10	NTP	114	NTP Version 4, server
3891	334.812247	192.168.121.40	212.227.54.68	NTP	94	NTP Version 3, client
3893	334.817994	212.227.54.68	192.168.121.40	NTP	94	NTP Version 3, server

We can see that the client(**2003:51:6012:121::10**) first sent a NTP packet(2918) to **2003:51:6012:110::dcf7:123**, and then received one back from **2003:51:6012:110::dcf7:123**(2919). which had the mode flag set to Server.

Since it is the only NTP client-server interaction over IPv6, I concluded that this was the answer.

Question 3

Answer: `ns1.hans.hosteurope.de`

First, I filtered by **DNS**.

dns

No.	Time	Source	Destination	Protocol	Length	Info
242	21.049259	192.168.121.2	192.168.120.22	DNS	82	Standard query 0xb4ca A blog.webernetz.net
243	21.050522	192.168.120.22	192.168.121.2	DNS	152	Standard query response 0xb4ca A blog.webernetz.net A 5.35.226.136 NS ns2.hans.hosteurope.de NS ns1.hans.hosteurope.de
851	81.049328	192.168.121.2	192.168.120.22	DNS	82	Standard query 0x3238 A blog.webernetz.net
913	84.047794	192.168.121.2	192.168.120.22	DNS	82	Standard query 0x3238 A blog.webernetz.net
939	87.047761	192.168.121.2	192.168.120.22	DNS	82	Standard query 0x3238 A blog.webernetz.net
966	90.047724	192.168.121.2	192.168.120.22	DNS	82	Standard query 0x3238 A blog.webernetz.net
1427	141.050146	192.168.121.2	192.168.120.22	DNS	82	Standard query 0xc1aa A blog.webernetz.net
1486	144.048490	192.168.121.2	192.168.120.22	DNS	82	Standard query 0xc1aa A blog.webernetz.net
1514	147.048589	192.168.121.2	192.168.120.22	DNS	82	Standard query 0xc1aa A blog.webernetz.net
1544	150.048552	192.168.121.2	192.168.120.22	DNS	82	Standard query 0xc1aa A blog.webernetz.net
2023	201.051215	192.168.121.2	192.168.120.22	DNS	82	Standard query 0x6306 A blog.webernetz.net
2091	204.049182	192.168.121.2	192.168.120.22	DNS	82	Standard query 0x6306 A blog.webernetz.net
2118	207.049168	192.168.121.2	192.168.120.22	DNS	82	Standard query 0x6306 A blog.webernetz.net
2154	210.050113	192.168.121.2	192.168.120.22	DNS	82	Standard query 0x6306 A blog.webernetz.net
2619	261.052035	192.168.121.2	192.168.120.22	DNS	82	Standard query 0x2aa5 A blog.webernetz.net
2620	261.053287	192.168.120.22	192.168.121.2	DNS	152	Standard query response 0x2aa5 A blog.webernetz.net A 5.35.226.136 NS ns1.hans.hosteurope.de NS ns2.hans.hosteurope.de
3506	321.053856	192.168.121.2	192.168.120.22	DNS	82	Standard query 0xe597 A blog.webernetz.net
3507	321.054857	192.168.120.22	192.168.121.2	DNS	152	Standard query response 0xe597 A blog.webernetz.net A 5.35.226.136 NS ns2.hans.hosteurope.de NS ns1.hans.hosteurope.de
3636	322.493083	2003:51:6012:121::2	2003:51:6012:120::a08:53	DNS	100	Standard query 0x6e4e AAAA ip.webernetz.net
3637	322.494205	2003:51:6012:120::a…	2003:51:6012:121::2	DNS	182	Standard query response 0x6e4e AAAA ip.webernetz.net AAAA 2003:51:6012:110::19 NS ns2.hans.hosteurope.de NS ns1.hans.hos

Then, looking at the responses, specifically I looked at packet 2620, I found the authoritative name servers for the domain that was queried(blog.webergetz.net), which were **ns1.hans.hosteurope.de** and **ns2.hans.hosteurope.de**

Question 4

Answer: `GigabitEthernet0/2`

I filtered by **CDP**, and the answer to the question can actually even be seen in the **Info column**: "*Device ID:CCNP-LAB-S2.webergetz.net **Port ID:GigabitEthernet0/2***"

cdp

No.	Time	Source	Destination	Protocol	Length	Info
133	11.088970	Cisco_a1:5a:9a	CDP/VTP/DTP/PAgP/UDLD	CDP	498	Device ID: CCNP-LAB-S2.webernetz.net Port ID: GigabitEthernet0/2
138	11.748317	Cisco_ae:31:99	CDP/VTP/DTP/PAgP/UDLD	CDP	541	Device ID: CCNP-LAB-S1.webernetz.net Port ID: GigabitEthernet0/1
761	71.098782	Cisco_a1:5a:9a	CDP/VTP/DTP/PAgP/UDLD	CDP	498	Device ID: CCNP-LAB-S2.webernetz.net Port ID: GigabitEthernet0/2
771	71.752519	Cisco_ae:31:99	CDP/VTP/DTP/PAgP/UDLD	CDP	541	Device ID: CCNP-LAB-S1.webernetz.net Port ID: GigabitEthernet0/1
1335	131.112111	Cisco_a1:5a:9a	CDP/VTP/DTP/PAgP/UDLD	CDP	498	Device ID: CCNP-LAB-S2.webernetz.net Port ID: GigabitEthernet0/2
1346	131.756713	Cisco_ae:31:99	CDP/VTP/DTP/PAgP/UDLD	CDP	541	Device ID: CCNP-LAB-S1.webernetz.net Port ID: GigabitEthernet0/1
1921	191.121684	Cisco_a1:5a:9a	CDP/VTP/DTP/PAgP/UDLD	CDP	498	Device ID: CCNP-LAB-S2.webernetz.net Port ID: GigabitEthernet0/2
1935	191.761025	Cisco_ae:31:99	CDP/VTP/DTP/PAgP/UDLD	CDP	541	Device ID: CCNP-LAB-S1.webernetz.net Port ID: GigabitEthernet0/1
2511	251.133254	Cisco_a1:5a:9a	CDP/VTP/DTP/PAgP/UDLD	CDP	498	Device ID: CCNP-LAB-S2.webernetz.net Port ID: GigabitEthernet0/2
2521	251.788355	Cisco_ae:31:99	CDP/VTP/DTP/PAgP/UDLD	CDP	541	Device ID: CCNP-LAB-S1.webernetz.net Port ID: GigabitEthernet0/1
3282	311.149194	Cisco_a1:5a:9a	CDP/VTP/DTP/PAgP/UDLD	CDP	498	Device ID: CCNP-LAB-S2.webernetz.net Port ID: GigabitEthernet0/2
3304	311.786290	Cisco_ae:31:99	CDP/VTP/DTP/PAgP/UDLD	CDP	541	Device ID: CCNP-LAB-S1.webernetz.net Port ID: GigabitEthernet0/1

Question 5

Answer: `Version 12.1(22)EA14`

After filtering by **CDP**, looking at any of the packets from CCNP-LAB-S2 (133 for example), we can see the version in the **Software Version** section of the packet:

```
- Software Version
    Type: Software version (0x0005)
    Length: 276
    Software version: Cisco Internetwork Operating System Software
    Software version: IOS (tm) C2950 Software (C2950-I6K2L2Q4-M), Version 12.1(22)EA14, RELEASE SOFTWARE (fc1)
    Software version: Technical Support: http://www.cisco.com/techsupport
    Software version: Copyright (c) 1986-2010 by cisco Systems, Inc.
    Software version: Compiled Tue 26-Oct-10 10:35 by nburra
- Platform: cisco WS-C2950G-24-EI

0090  6d 20 53 6f 66 74 77 61  72 65 20 0a 49 4f 53 20   m Softwa re ·IOS
00a0  28 74 6d 29 20 43 32 39  35 30 20 53 6f 66 74 77   (tm) C29 50 Softw
00b0  61 72 65 20 28 43 32 39  35 30 2d 49 36 4b 32 4c   are (C29 50-I6K2L
00c0  32 51 34 2d 4d 29 2c 20  56 65 72 73 69 6f 6e 20   2Q4-M),  Version
00d0  31 32 2e 31 28 32 32 29  45 41 31 34 2c 20 52 45   12.1(22) EA14, RE
00e0  4c 45 41 53 45 20 53 4f  46 54 57 41 52 45 20 28   LEASE SO FTWARE (
00f0  66 63 31 29 0a 54 65 63  68 6e 69 63 61 6c 20 53   fc1)·Tec hnical S
0100  75 70 70 6f 72 74 3a 20  68 74 74 70 3a 2f 2f 77   upport:  http://w
0110  77 77 2e 63 69 73 63 6f  2e 63 6f 6d 2f 74 65 63   ww.cisco .com/tec
0120  68 73 75 70 70 6f 72 74  0a 43 6f 70 79 72 69 67   hsupport ·Copyrig
```

Question 6

Answer: `21:02:36 UTC on Friday, March 3 2017`

I found this by trying to look up through which protocols data about the NVRAM config would be sent, which brought me to [this link], which mentioned **FTP/TFTP**

I also found some stackoverflow questions online that where I saw that this information looks roughly like *"NVRAM config last updated at..."*

So I first tried to filter by FTP, found no packets, then I filtered by TFTP and saw there were multiple packets, so I finally used this filter: **tftp contains "NVRAM config"** and found packet 3770. This is the packet's data as printable text(minus the noise at the beginning), which contains the answer:

```
!
! Last configuration change at 20:55:45 UTC Fri Mar 3 2017 by
weberjoh
! NVRAM config last updated at 21:02:36 UTC Fri Mar 3 2017 by
weberjoh
! NVRAM config last updated at 21:02:36 UTC Fri Mar 3 2017 by
weberjoh
version 15.1
service timestamps debug datetime msec
service timestamps log datetime msec
service password-encryption
!
hostname CCNP-LAB-R2
!
boot-start-marker
boot-end-marker
!
!
enable secret 5 $1$Z.9j$Nvobsx9NvJzqtRLQqR.9b0
!
aaa new-model
!
!
aaa group server radius foobar
 server name blubb
```